

Exercise 3.2. Let G be a group and let $H \subset G$ be a subgroup.

- Show that $g_1 \sim g_2 :\Leftrightarrow g_1^{-1}g_2 \in H$ defines an equivalence relation on G .
- Use part (a) of this exercise to show that any two cosets in G are either equal or disjoint.

You can try to [verify part \(a\) of this exercise with Lean](#).

Hints for Lean

```
import Mathlib.Algebra.Group.Subgroup.Basic

variable {G : Type} [Group G] (H : Subgroup G)

def Rel (H : Subgroup G) (g g' : G) : Prop := g⁻¹ * g' ∈ H

lemma Rel_is_equivalence_relation : Equivalence (Rel H) := by
  constructor
  · sorry
  · sorry
  · sorry
```

A relation R on a set M is usually defined as a subset of $M \times M$. In Lean, however, a relation on M is defined as a function $M \rightarrow M \rightarrow \text{Prop}$. The idea is that for each $x, y \in M$ we obtain a proposition $P(x, y)$ which is `true` precisely when $(x, y) \in R$. In our example, the function `Rel` sends $g \in G$ to the function that sends $g' \in G$ to the statement $g^{-1} \cdot g' \in H$. You can read more about this on

https://leanprover-community.github.io/logic_and_proof/relations_in_lean.html.

The `constructor` tactic splits the current goal into three subgoals, one for each defining property of an equivalence relation. The bullet points are syntax to separate the three subgoals.

A [solution](#) is available on the repository

<https://github.com/pedro-nunez/lean-algebra-2021>.

Lean's search tactics

Many simple statements are already implemented in `mathlib`. The `exact?` tactic searches for results already proven in `mathlib` that can help solve the current goal. For the first `sorry` above, try the following:

```
intro g
unfold Rel
simp only [inv_mul_cancel]
exact?
```

Other search tactics you can try are `apply?`, `simp?` and `rw?`.

[†]Exercises designed for the lecture *Algebra and Number Theory*, for which I was a teaching assistant. This was an introductory lecture on abstract algebra, polynomial rings and Galois theory, taught by Annette Huber-Klawitter at the University of Freiburg during the Winter Semester 2021/2022.